

CustomerIQ - Final Tech Stack

Frontend

React TypeScript Vite

UI Components

shadcn/ui

Styling

Tailwind CSS

Animations

Framer Motion

Charts

Recharts

Icons

Lucide React

Backend

FastAPI

Data Analytics

Pandas NumPy

Authentication

JWT Authentication

Database

PostgreSQL 17

ORM

SQLAlchemy

Database Migrations

Alembic

Validation

Pydantic

Reports

ReportLab

File Processing

OpenPyXL CSV

Deployment

Frontend: Vercel

Backend: Render

Database: Neon PostgreSQL

AI Layer

Rule-Based Recommendation Engine

(No paid APIs required)

Version Control

Git GitHub